```
TensorFlow: 2.19.1
Working from: /Users/eloelo/Projects/plant-disease-detection

Test images: 1825
Test batches: 58

All 4 models loaded

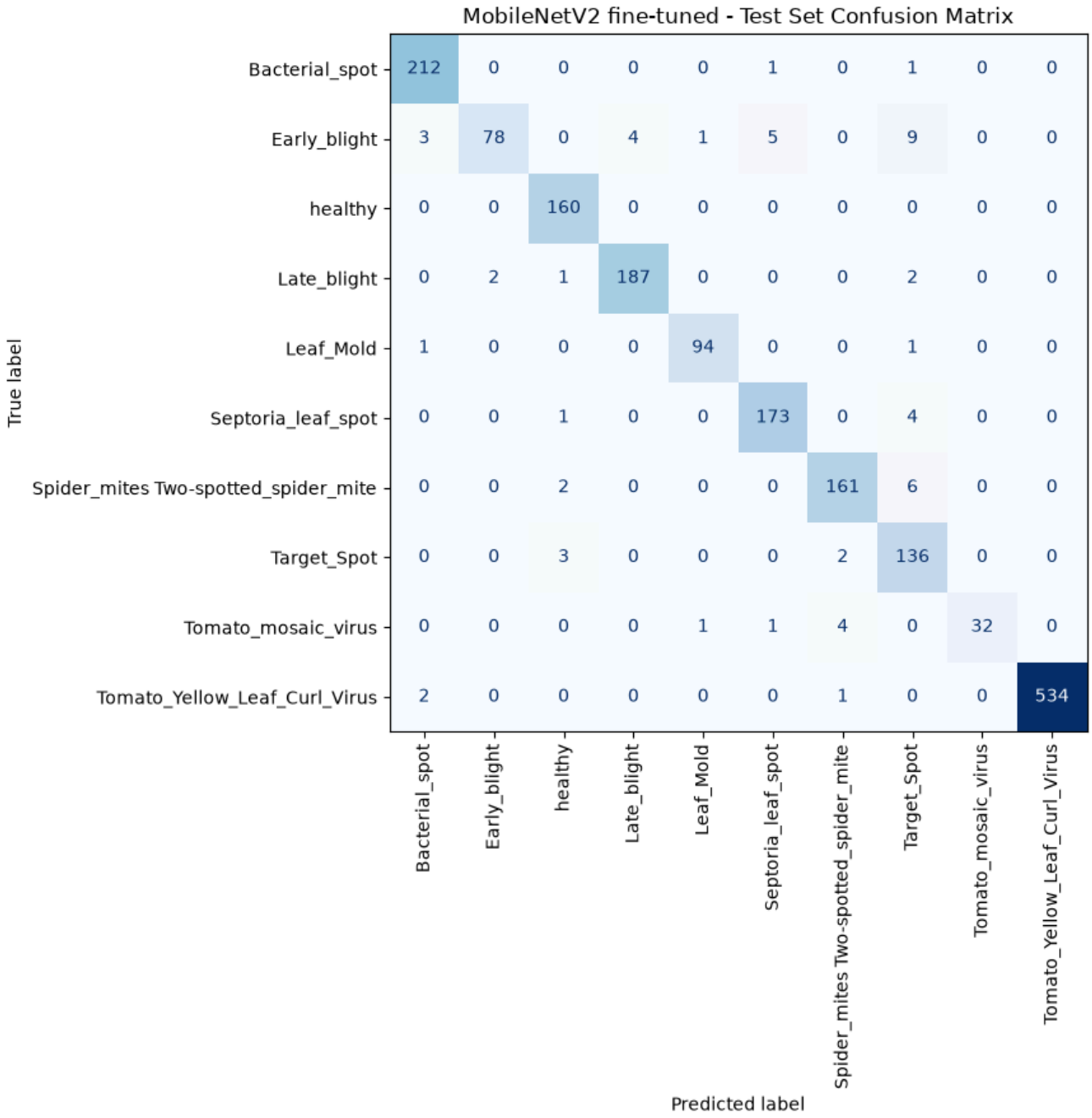
Baseline CNN
  Test accuracy: 90.19 %
  Macro-F1:      0.8665
MobileNetV2 frozen
  Test accuracy: 91.51 %
  Macro-F1:      0.8818
MobileNetV2 fine-tuned
  Test accuracy: 96.82 %
  Macro-F1:      0.9523
ResNet50 frozen
  Test accuracy: 96.66 %
  Macro-F1:      0.958
```

MobileNetV2 fine-tuned

	precision	recall	f1-score	support
Bacterial_spot	0.972	0.991	0.981	214
Early_blight	0.975	0.780	0.867	100
healthy	0.958	1.000	0.979	160
Late_blight	0.979	0.974	0.977	192
Leaf_Mold	0.979	0.979	0.979	96
Septoria_leaf_spot	0.961	0.972	0.966	178
Spider_mites Two-spotted_spider_mite	0.958	0.953	0.955	169
Target_Spot	0.855	0.965	0.907	141
Tomato_mosaic_virus	1.000	0.842	0.914	38
Tomato_Yellow_Leaf_Curl_Virus	1.000	0.994	0.997	537
accuracy			0.968	1825
macro avg	0.964	0.945	0.952	1825
weighted avg	0.970	0.968	0.968	1825

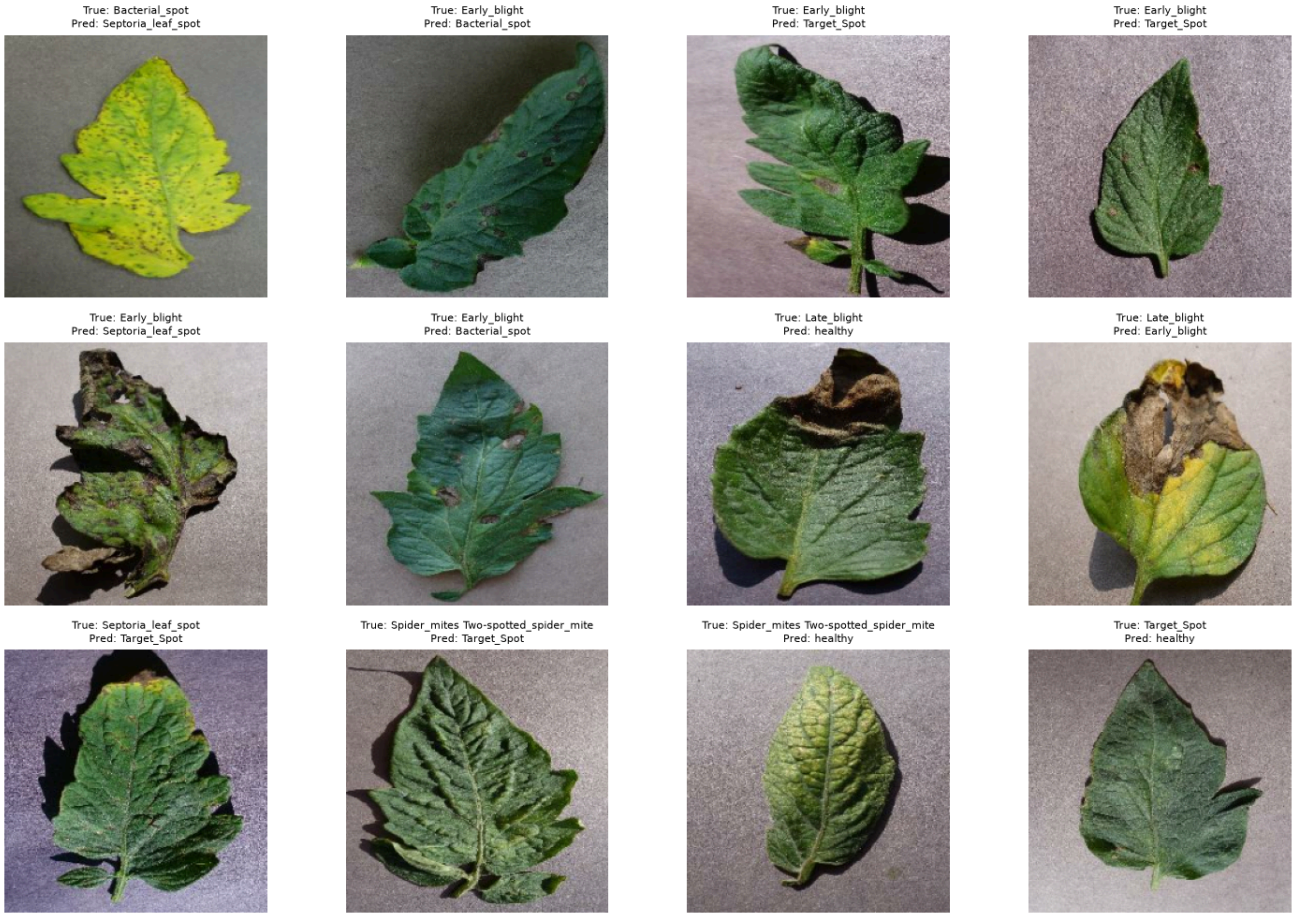
ResNet50 frozen

	precision	recall	f1-score	support
Bacterial_spot	0.981	0.963	0.972	214
Early_blight	0.977	0.860	0.915	100
healthy	0.925	1.000	0.961	160
Late_blight	0.969	0.979	0.974	192
Leaf_Mold	1.000	0.948	0.973	96
Septoria_leaf_spot	0.932	1.000	0.965	178
Spider_mites Two-spotted_spider_mite	0.942	0.959	0.950	169
Target_Spot	0.913	0.894	0.903	141
Tomato_mosaic_virus	0.974	0.974	0.974	38
Tomato_Yellow_Leaf_Curl_Virus	1.000	0.987	0.993	537
accuracy			0.967	1825
macro avg	0.961	0.956	0.958	1825
weighted avg	0.967	0.967	0.966	1825


MobileNetV2 fine-tuned - Test Set Confusion Matrix
True label
Predicted label
Bacterial_spot
Early_blight
healthy
Late_blight
Leaf_Mold
Septoria_leaf_spot
Spider_mites Two-spotted_spider_mite
Target_Spot
Tomato_mosaic_virus
Tomato_Yellow_Leaf_Curl_Virus
212 0 0 0 0 1 0 1 0 0
3 78 0 4 1 5 0 9 0 0
0 0 160 0 0 0 0 0 0 0
0 2 1 187 0 0 0 2 0 0
1 0 0 0 94 0 0 1 0 0
0 0 1 0 0 173 0 4 0 0
0 0 2 0 0 0 161 6 0 0
0 0 3 0 0 0 2 136 0 0
0 0 0 0 1 1 4 0 32 0
2 0 0 0 0 0 1 0 0 534




```
Wrong predictions: 58
```



Test set results

- Test set used only here, never for training or choosing models
- Test results within about 1 point of val for every model, so the models generalise
- MobileNetV2 fine-tuned 96.8% acc (macro-F1 0.952), ResNet50 frozen 96.7% (0.958): basically tied
- Baseline 90.2% (0.867): biggest gap between accuracy and macro-F1, so weakest on small classes
- MobileNetV2 hardest class: Early_blight (78 of 100), mostly mistaken for Target_Spot
- Mosaic virus: 32 of 38, the 6 missed went to other diseases, never to healthy
- Diseased leaves called healthy: MobileNetV2 7, ResNet50 13
- Most errors are look-alike diseases or leaves with faint symptoms, but one Late_blight leaf with a large dead patch at the edge was called healthy, so the model can miss damage that covers only part of the leaf
- Chosen for the app: MobileNetV2 fine-tuned, as accurate, 10x smaller, and fewer sick called healthy mistakes

```
Baseline CNN
  Top-3 accuracy: 98.52 %
  Latency per image: 23.8 ms
```

```
MobileNetV2 frozen
  Top-3 accuracy: 99.01 %
  Latency per image: 36.9 ms
MobileNetV2 fine-tuned
  Top-3 accuracy: 99.73 %
  Latency per image: 36.6 ms
ResNet50 frozen
  Top-3 accuracy: 99.73 %
  Latency per image: 36.2 ms
```

Summary of test results

Model	Accuracy	Macro-F1	Top-3 accuracy	Latency per image
Baseline CNN	90.2%	0.867	98.5%	23.8 ms
MobileNetV2 frozen	91.5%	0.882	99.0%	36.9 ms
MobileNetV2 fine-tuned	96.8%	0.952	99.7%	36.6 ms
ResNet50 frozen	96.7%	0.958	99.7%	36.2 ms

- Top-3 accuracy 99.7% for the two best models: the right class is almost always in the top 3, so the app shows the top 3
- MobileNetV2 and ResNet50 take about the same time, so I picked MobileNetV2 for its size (2.3M vs 23.6M parameters), not speed

Is the model usable for the target users?

- Yes, as a first screening tool for farm advisors and field workers: 96.8% accuracy, 99.7% top-3, instant answers
- Only for tomato leaves photographed like the dataset: one leaf, plain background
- Not tested on field photos with soil, other leaves or poor light, so accuracy there is unknown
- Weakest on Early_blight (78% found), often mistaken for Target_Spot
- 7 of the 1,665 diseased test leaves were called healthy, so a "healthy" result should not stop an expert check
- Advisory and educational use only, not a final diagnosis; the app will show confidence scores and these limitations